Week 06: Proposal Writing I: Objectives, Originality, Method and Work Plan

R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

A research proposal is an argument that a problem matters, that the proposed work will solve part of it and that the team can deliver. This week covers the core sections: objectives, originality, method, work plan and the pathway to impact [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to write objectives that are specific, measurable and time-bound, and to state originality against the literature. They should be able to align the method and work packages with the objectives and to plan risks with alternatives. Finally, they should be able to describe impact with a logic model.

Lecture

The proposal as an argument

Every funding scheme asks the same questions in different words. Is the problem important and the idea original? Will the method reach the objectives? Can the team deliver on time and within budget? Will the results matter beyond the project? TÜBİTAK's form is organised around originality, method, project management and wider impact [3], and Horizon Europe's around excellence, impact and implementation [4]. A strong proposal answers these questions consistently: Each objective has a method, each method has tasks, each task has resources, and each result has a route to use. Reviewers notice inconsistencies between sections quickly, because they read with the criteria at hand.

Objectives and originality

Doran proposed that objectives should be specific, measurable, assignable, realistic and time-related, the origin of the acronym SMART [1]. In research, objectives should be verifiable at the end of the project. An objective such as improving detection is not verifiable. Reducing the false

negative rate of a detector on a public benchmark from a stated baseline within eighteen months is verifiable. Research questions and hypotheses make the objectives testable. Originality is stated against the literature: what is known, what is missing, and precisely what the project adds, whether a new method, new data, new theory or a new application. The review protocol of Week 3 provides the evidence.

Method and work plan

The method explains how each objective will be reached, including the design of experiments from Week 2. The work plan turns the method into work packages, tasks, deliverables and milestones on a timeline. A deliverable is a tangible output, such as a report, a dataset or a software release, while a milestone is a checkpoint at which progress is verified. TÜBİTAK's form asks for a plan B for the main risks, which must keep the project on its core objectives [3]. Risks are covered in depth in Week 10, and scheduling in Week 9.

The pathway to impact

Impact is the most often underdeveloped section. A logic model makes the pathway explicit: inputs enable activities, activities produce outputs, outputs lead to outcomes and outcomes contribute to long-term impact [2]. Figure 6.1 shows an example. Outputs, such as papers and software, are under the project's control. Outcomes, such as changes in how hospitals or developers work, are influenced but not controlled, and impact appears over years. Horizon Europe distinguishes three related activities: Dissemination makes results available to those who can use them, exploitation uses them, and communication informs the wider public. A credible impact section names the users, the barriers to uptake and the measures that will show progress.

> ***Pause and reflect.*** *Which of your objectives would a reviewer find hardest to verify at the end of the project?*

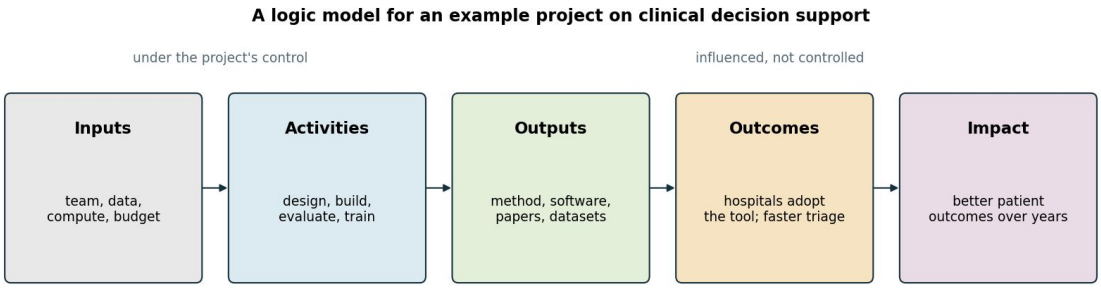


Figure 6.1. A logic model: inputs and activities lead to outputs under the project's control, and to outcomes and impact that the project influences [2].

Hands-on components

Interactive lab: Part A checks draft objectives for measurable targets, deadlines and vague verbs. Part B builds a logic model. Part C draws work packages as a timeline. The checks are

heuristics that support, not replace, careful reading [1, 2]. Link: https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-06/lab.html

Colab notebook: It implements simple checks for draft objectives and applies them to a set of examples, to show what such checks can and cannot detect [1]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-06/NB06_proposal_writing_1.ipynb

Reflection prompts

1. Rewrite your weakest objective until it passes the checks. What did you have to decide in order to make it measurable?
2. Which outcome in your logic model depends most on people outside the project? How will you influence it?
3. Where does your work plan depend on a single task that could fail? What is the plan B?

Weekly task and submission

What to do. Write three sections of your proposal, about 900 words in total: the objectives, the originality and the work plan. State each objective so that it is specific, measurable and time-bound. Position the originality against the literature of your Week 3 review. Present the work plan as a table of work packages with a timeline figure. Add a logic model that leads from activities to outputs, outcomes and impact.

Where to look. Lecture: sections "Objectives and originality", "Method and work plan" and "The pathway to impact". Interactive lab: Part A for checking objectives, Part B for the logic model and Part C for the timeline of work packages. Colab notebook: Section 1 with both exercises, which check draft objectives.

Example. The objective "Improve the detection of hate speech" cannot be verified at the end of the project. A verifiable version is "Reduce the false negative rate of the detector on a public benchmark from 18 percent to below 12 percent by month 18".

What to submit. About 900 words with the work package table, the timeline figure from the interactive lab and the logic model, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Doran, G. T. (1981). There's a S.M.A.R.T. way to write management's goals and objectives. Management Review, 70(11), 35-36.

[2] W. K. Kellogg Foundation (2004). Logic Model Development Guide. W. K. Kellogg Foundation.

[3] TÜBİTAK (2026). 1001 Bilimsel ve Teknolojik Araştırma Projelerini Destekleme Programı: Proje başvuru formu. https://tubitak.gov.tr/sites/default/files/20689/1001_basvuru_formu.doc

[4] European Commission (2021). Standard briefing slides for experts: Horizon Europe. https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/experts/standard-briefing-slides-for-experts_he_en.pdf